

CHEIRAP (■■■■■) — Official Pitching Script & Jury Defense

National Innovation Challenge on AI & Digital Governance – Manipur (2026) | Host: DIT Manipur & MTIF

Problem Track: PWD-04	Category: AI-Based Construction Progress Monitoring (Govt-side)	Total Time: 6 Minutes (Strict)
Submitted By: Team Lotux	Target Dept: Public Works Department (PWD), Govt of Manipur	Structure: 4m Pitch + 2m Q&A;

OFFICIAL JURY INSTRUCTIONS: Total 6-minute slot (4 min pitch + 2 min Q&A;). Firm buzzer. Keep words simple, clear, and confident (~130 words/min). The live prototype dashboard is your primary truth.

MINUTE 1: The Problem & The Trap of Generic AI (0:00 – 1:00)

[STAGE & SCREEN ACTION — 0:00 – 1:00]: Stand tall, look at the judges. Keep the CHEIRAP surveillance hero dashboard visible on screen.

"Good morning, respected jury members. I am representing Team Lotux. Today, we are presenting our solution for Problem PWD-04: AI-Based Construction Progress Monitoring for the Public Works Department.

Across Manipur, the government manages hundreds of capital works—schools, bridges, roads, and health centers—frequently across remote hill districts. The problem stated in the hackathon manual is simple: field engineers cannot physically travel to remote sites every week to verify physical progress.

Because of this, the department relies on self-reported photographs and paper bills. What is the result? By the time an audit team visits two years later, public funds have already been disbursed for works that were delayed, sub-standard, or never built at all.

Now, when other teams approach this problem, they propose a generic AI model that looks for bricks or workers. But in real government administration, that fails. Fraudulent contractors do not upload random pictures; they use **photo recycling**—reusing real photos of an older school built two years ago in another district. A basic vision model sees bricks and marks it verified. We knew we had to solve the real problem."

Target: 60s | Spoken Words: ~175 | Tone: Crisp, conversational, plain English

MINUTE 2: The Core Innovation — The 4 Verification Pillars (1:00 – 2:00)

[STAGE & SCREEN ACTION — 1:00 – 2:00]: Step toward the screen. Click on the 'Works Assurance' tab showing the 4 verification pillars.

"That is why we built **CHEIRAP**. CHEIRAP does not guess; it validates physical works across four real-world engineering checks:

First, Photo Forensics: We calculate a 64-bit digital fingerprint called a perceptual hash for every uploaded photo. We cross-check that fingerprint against our statewide archive of over 11,000 civil works to instantly catch recycled imagery.

Second, Real Geofencing: We extract hardware GPS coordinates and reverse-geocode them against the sanctioned survey boundary polygon. If a photo was taken outside the approved boundary, it is immediately flagged.

Third, Engineering Physics: Concrete takes time to cure. Construction cannot jump from 15% to 100% in 12 days during peak monsoon. CHEIRAP compares progress against standard CPWD velocity curves.

Fourth, Fiscal Parity: We match Running Account bills against physical Measurement Book entries signed by the field engineer. If someone claims full payment without signed measurements, it is blocked."

Target: 60s | Spoken Words: ~150 | Tone: Crisp, conversational, plain English

MINUTE 3: Live System Demo — Case A vs. Case B (2:00 – 3:00)

[STAGE & SCREEN ACTION — 2:00 – 3:00]: Open Case A in the camera viewport modal. Point to HUD reticles and red 93.4% match banner, then switch to Case B.

"Let us see this working live on our prototype right now.

Look at **Case A** on the screen. The contractor submitted this photo claiming progress on a school in Lamphelpat. CHEIRAP instantly caught two fatal anomalies: First, our perceptual hash found a **93.4% duplicate match** with an archived 2024 project in Bishnupur. It caught the recycled photo on the spot. Second, the GPS coordinates revealed the camera was **9.4 kilometers away** from the sanctioned boundary. The running bill was frozen immediately.

Now look at **Case B**—Churachandpur Model College. Over 45 days, the imagery shows the civil frame progressing to lab installation. The photos are 100% unique, inside the geofence, and consistent with CPWD curing timelines. CHEIRAP awards it a **Green Passport clearance**, so honest contractors get paid on time without bureaucratic delays."

Target: 60s | Spoken Words: ~145 | Tone: Crisp, conversational, plain English

MINUTE 4: Why CHEIRAP Wins & Closing Punchline (3:00 – 4:00)

[STAGE & SCREEN ACTION — 3:00 – 4:00]: Face the judges directly. Show the generated PWD Form 44 and Section 65B Certificate on the screen.

"What makes our proposal truly unique compared to other solutions? Three fundamental things:

1. **Administrative Due Process:** Other AI tools act like an arbitrary black box that bans contractors—which gets stayed in court the next day. CHEIRAP respects natural justice. When an anomaly is detected, it automatically issues an official **PWD Form 44 Field Inspection Notice**, giving the Superintending Engineer a mandatory 7-day window to physically verify the site.
2. **Court-Ready Evidence:** Under **Section 65B of the Bharatiya Sakshya Adhiniyam, 2023**, every photo, GPS coordinate, and hash check is sealed with an automated digital certificate. If a case goes to the High Court of Manipur or Lokayukta, the evidence is legally admissible from day one.
3. **Zero Workflow Disruption:** CHEIRAP plugs directly into existing government systems like Manipur Works MIS and GePNIC without replacing them.

To conclude: Integrity in public construction should not be an autopsy performed two years after funds are lost. With CHEIRAP, the Public Works Department gets an automated, honest, 24/7 gatekeeper. Thank you. We are ready for your questions."

Target: 60s | Spoken Words: ~165 | Tone: Crisp, conversational, plain English

PITCH EXECUTION TIMELINE & BUZZER MANAGEMENT CHEATSHEET:

- **0:00 – 1:00:** Hook the jury with the real hill-district problem and expose why basic vision models fail.
- **1:00 – 2:00:** Introduce CHEIRAP's 4 checks (pHash, Boundary Geofence, CPWD Curves, Measurement Book).
- **2:00 – 3:00:** Live Demo! Contrast Case A (Red Flag: 93.4% duplicate photo) against Case B (Green Passport).
- **3:00 – 3:50:** Deliver the 3 differentiators (Due Process, Section 65B BSA Seals, Zero Disruption) + Punchline.
- **3:50 – 4:00:** Transition cleanly to Q&A; before the 4-minute warning bell rings.

The 2-Minute Q&A; Session: Trilateral Jury Defense Cards

Direct, authoritative 30-second responses mapped to the official hackathon evaluation criteria.

PILLAR 1: TECHNICAL TRUST (35% WEIGHTAGE) — Anticipated Jury Question

Judge: "What happens if a contractor tampers with the EXIF metadata or changes the phone's GPS coordinates before uploading?"

Core Strategy: Demonstrate multi-vector redundancy. Show that spoofing text cannot alter the visual scene.

Your 30-Second Spoken Answer: "That is exactly why CHEIRAP does not rely on GPS alone. GPS is only Vector 1. Even if someone tampers with the EXIF metadata to fake coordinates in Lamphelpat, our 64-bit perceptual hash immediately compares the actual image against our 11,202 civil work database. You can edit text in an image file, but you cannot fake the background pixels. Furthermore, any altered file structure fails our Section 65B SHA-256 cryptographic integrity check."

PILLAR 2: GOVERNMENT RELEVANCE (30% WEIGHTAGE) — Anticipated Jury Question

Judge: "How does this function in remote hill districts like Tamenglong or Pherzawl where mobile internet connectivity is completely unavailable?"

Core Strategy: Emphasize offline-first field capture and asynchronous synchronization.

Your 30-Second Spoken Answer: "CHEIRAP is designed specifically for Manipur's geography. Field engineers and supervisors use their standard mobile camera offline. The device's GPS satellite receiver and hardware timestamp operate independently of cellular networks, embedding verifiable EXIF coordinates into the image. When the officer returns to the block development office or connects to Wi-Fi, the files batch-upload to CHEIRAP for hashing and verification. The workflow never halts due to lack of network."

PILLAR 3: INDUSTRY POTENTIAL (35% WEIGHTAGE) — Anticipated Jury Question

Judge: "Can this system scale across the entire state, and what is the computational cost of running perceptual hashes across thousands of images?"

Core Strategy: Highlight algorithmic efficiency: DCT pHash is sub-millisecond and runs on standard commodity servers.

Your 30-Second Spoken Answer: "Perceptual hashing using 64-bit Discrete Cosine Transform is mathematically lightweight. Comparing two 64-bit integer hashes via Hamming distance takes microseconds in CPU cache. Our FastAPI microservice processes over 500 site photographs per second on a single commodity server. Across all 16 districts of Manipur, annual photo submissions number under 50,000, requiring virtually zero expensive GPU infrastructure. The same architecture easily scales nationwide to all 28 states."

BONUS: ADMINISTRATIVE LAW & LEGAL TRUST (30% WEIGHTAGE) — Anticipated Jury Question

Judge: "Why issue a PWD Form 44 notice instead of immediately blacklisting the contractor upon detecting an anomaly?"

Core Strategy: Direct alignment with administrative law, Article 14, and avoiding judicial stay orders.

Your 30-Second Spoken Answer: "If an algorithm unilaterally blacklists a contractor without a hearing, the High Court will stay the order within 24 hours under the doctrine of natural justice. CHEIRAP protects the government legally. By freezing running disbursements while automatically issuing a PWD Form 44 notice with a 7-day physical verification window, we uphold administrative due process. The state prevents fund leakage, while the engineer has statutory backing to conduct an on-site audit."

ON-SITE REGISTRATION & SETUP REMINDER (MORNING OF 9 OCT 2026):

Registration opens at **9:00 AM** sharp. Arrive early, verify laptop HDMI/DisplayPort output, test the local web server at <http://localhost:5173>, and confirm that both Case A and Case B modals load smoothly offline.